

Abhinav Rawat

B.Tech + M.Tech - Electronics & Communication Engineering
The LNM Institute of Information Technology, Jaipur
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BRIEF SUMMARY

Electronics & Communication Engineer specializing in Wireless Systems (PHY/MAC layers), FPGA-based hardware prototyping (VHDL), and signal processing. Experienced in building embedded transceivers, channel modeling, and resource optimization for next-generation communication networks.

KEY EXPERTISE

Skills: MATLAB, Python, 5G PHY, 3Gpp LaTeX, FPGA (VHDL, Basys 3), Analog Circuitry, Signal Processing, R&D, Circuit Design

EDUCATION

The LNM Institute of Information Technology Jaipur, 2021–2026
B.Tech + M.Tech - ECE
Cumulative CPI: 7.10/10 (71%)
Thesis: Efficient Resource Optimization for Indoor VLC Networks
Global Agra Convent School Agra, 2020
12th Grade (CBSE); Percentage: 83.40%
St. George's College Agra, 2018
10th Grade (CISCE); Percentage: 89.40%

PROFESSIONAL EXPERIENCE

ArcelorMittal Design and Engineering Centre Surat, India
Trainee– Full Time Jul 2026–Present
• Currently working on automating process in Industrial Steel Fabrication plants.

IIIT Delhi — Core Engineering New Delhi
Summer Research Intern May 2024–Aug 2024
• Key Skills: MATLAB, Signal Processing, Mobility Modeling
• Investigated hybrid Li-Fi/Wi-Fi environments and simulated user mobility to calculate SINR, handover rates, and throughput.
• Modeled indoor blockages (FixOM/SAM) and analyzed their impact on Li-Fi channel performance.
• Analyzed the effect of User Equipment (UE) orientation on system SINR.
• Optimized multi-user mobility handover algorithms to ensure seamless connectivity and optimal data rates.

DNN Wireless Technologies — Core Engineering Remote/On-site
Research Fellow Apr 2023–Mar 2024
• Key Skills: FPGA (VHDL), Analog Circuitry, Prototyping
• Prototyped three distinct Li-Fi/VLC systems, including receiver circuitry design.
• Developed an FPGA-based full-duplex IR Li-Fi system for multimedia transmission.
• Implemented a cooperative Li-Fi model to mitigate NLOS issues, integrating optical lenses for extended range.
• Engineered a wide-FOV LED/Solar-cell VLC system for broadcasting; achieved 1.6 Mbps data rate using FPGA decoding.

PROJECTS

LED-Solar Cell based VLC Downlink System (Broadcasting) Dec 2023–Mar 2024
Designed a wide-FOV VLC broadcasting system using LEDs and Solar cells. Developed noise-immune receiver circuitry and implemented FPGA-based decoding to achieve 1.6 Mbps.

Li-Fi Cooperative Communication Model Jul 2023–Aug 2023
Designed an IR-based cooperative Li-Fi system to overcome NLOS challenges. Incorporated optical lenses to enable long-distance multimedia transmission.

Full Duplex IR Li-Fi System Apr 2023–Jul 2023
Programmed an FPGA-based full-duplex Li-Fi system using IR sensors. Achieved a transmission range of 6 feet

for multimedia signals without data rate compromise.

PUBLICATIONS & RESEARCH

Layer-2 Hierarchical RSMA for Rate Fairness in Indoor Multi-AP VLC Networks 2026

Submitted to IEEE Internet of Things (IoT) Journal – Under Review

Formulated mathematical models for a Layer-2 Rate-Splitting Multiple Access (RSMA) framework, optimizing resource allocation and Successive Interference Cancellation (SIC) to improve user rate fairness.

Joint LED Placement and CoMP Association Optimization in Indoor VLC Systems 2026

Submitted to IEEE Networking Letters – Under Review

Investigated the joint scheduling and placement optimization of LED access points for coordinated multipoint joint transmission (CoMP-JT) in indoor visible light communication (VLC) to optimize system throughput.

Adaptive Link Selection and Handover in Orientation-Aware Hybrid Li-Fi/Wi-Fi Networks 2026

Journal: IEEE Networking Letters.

Investigated link selection mechanisms and handover optimization in hybrid optical/RF networks, incorporating user equipment orientation.

OAM-LIFINet: Orientation Aware Multi-AP Li-Fi Network with User Mobility Nov 2025

Conference: IEEE Future Network World Forum. — DOI: 10.1109/FNWF66845.2025.11317169.

Proposed the effect of device/UE orientation using Orientation based Random Waypoint Mobility Model (ORWP) on user throughput and interference mitigation.

FixOM-SAM: Blockage Analysis in Indoor Li-Fi Networks Jul 2025

Journal: IEEE Open Journal of the Communications Society (Q1) — DOI: 10.1109/OJCOMS.2025.3585614.

Proposed 2 different approaches to model static and human blockages in indoor Li-Fi Networks and analysed the effect of these blockages on system data rates.

Implementation of Cooperative Relaying based Li-Fi Infrared Communication Mar 2024

Conference: ISRO's RTST Conference 2024. Demonstrated FPGA-based Cooperative Li-Fi to enhance reliability in NLOS scenarios.

SEMINARS & WORKSHOPS

Bridging Digital Divide: MA Techniques for Aerospace Networks, IIIT Delhi Mar 2025

Unlocking the Potential of 6G, IIIT Delhi Jul–Oct 2024

ISRO RTST Conference 2024, ISRO Mar 2024

India Mobile Congress 2023, Dept. of Telecommunication Oct 2023

PERSONAL INTERESTS

Interests: Rock music, Chess, Travelling

Languages: English, Hindi